(a) SSM backbone

Aligned EEG $\tilde{X}$ — C × T

Multi-scale temporal
- Conv 1×20 BN
- Conv 1×32 BN
- Conv 1×64 BN

96 × C × T

Spatial filtering
- Depthwise conv C×1
- BN · ELU
- AvgPool 1×8

48 × T/8

Grouped temporal
- Grouped conv 1×16
- BN · ELU
- Group attention
- AvgPool 1×7

48 × T/56

Selective SSM encoder (×N)
- 1×1 conv · SiLU (mix)
- LayerNorm
- Linear
- Depthwise conv k=3
- Selective scan
- ⊙ SiLU gate
- Linear
- 1×1 conv · SiLU (reduce)

16 × T/56

64 × T/56

Temporal head
- Residual causal conv d=1
- Residual causal conv d=2

L_{tMMD}

f 64-dim feature

(b) Adaptation head

f_s , f_t — 64 = 4 groups × 16 channels

AutoDIAL normalization
- Batch moments: source, target
- Mix: α · target + (1 − α) · source
- Standardize per domain
- Shared affine w, b

Residual aligner
- LayerNorm
- Linear 64 → 128
- GELU · Dropout
- Linear 128 → 64
- LayerNorm
- × s (learned scale)

f'_s, f'_t

Classifier
- Grouped 1×1 conv (4 × K)
- Weighted group sum

$\hat{y}$

GRL

Discriminator
- Linear 64→128
- Linear 128→64
- Linear 64→1

L_{MMD} L_{CE} L_{IM} L_{adv}

Stage-1 objective

$$L = L_{CE} + \lambda_{IM}L_{IM} + \lambda_{adv}L_{adv} + \lambda_{MMD}L_{MMD} + \lambda_{t}L_{tMMD}$$

$\lambda_{IM} = 0.1$, $\lambda_{adv} = 1$, $\lambda_{MMD} = 0.5$, $\lambda_{t} = 0.1$

Two-stage adaptation

source · target · loss · TTA update

Stage 1 · training

X_s, y_s and X_t → Riemannian alignment → Backbone (a) all weights trained → Adaptation head (b) all weights and α → $\hat{y}$ → L_{total}

update

weights

Stage 2 · IM-TTA

X_t → Riemannian alignment → Backbone (a) only BN γ, β updated → Adaptation head (b) AutoDIAL stats re-estimated → $\hat{y}_t$ → L_{IM}

×5 steps